

Navigators Experimental Methodology

Data Normalization, Sequence Construction, Model Lifecycles & Ablation Framework

Navigators : Experimental Methodology & Ablation Protocol

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1. Experimental Design Philosophy

Navigators adheres to strict empirical guidelines for navigation benchmarking:

- Zero Ground-Truth Leakage:** During simulated or real GNSS outage intervals, GNSS measurements (including derived GNSS speed and ground-track heading) are completely withheld from the estimator. GNSS coordinates are sequestered strictly as an evaluation reference to compute positional error post-run.
- Identical Inputs Across Ablations:** To isolate the exact marginal utility of each algorithmic component, ablations A through G are evaluated on the exact same sensor stream, calibration prefix, and outage window.
- Deterministic Reproducibility:** Random seeds, filter tuning matrices, normalization parameters, and input dataset hashes are recorded in test logs.

2. Seven-Stage Ablation Hierarchy (A through G)

The evaluation suite (*replay.py* and *scripts/benchmark.py*) defines seven progressive system configurations:

Mode	Identifier	Filtering	AI Velocity	15-State EKF	NHC Constraints	ZUPT Detector	Map Matching
A	Raw IMU Baseline	■	■	■	■	■	■
B	Filtered IMU	■	■	■	■	■	■
C	AI Velocity Only	■	■	■	■	■	■
D	AI + EKF Fusion	■	■	■	■	■	■
E	AI + EKF + NHC	■	■	■	■	■	■
F	AI + EKF + NHC + ZUPT	■	■	■	■	■	■
G	Full Production System	■	■	■	■	■	■

Architectural Roles of Each Stage:

- Mode A (Raw IMU):** Direct double-integration of aligned raw accelerometer readings. Illustrates unconstrained MEMS drift where quadratic position error accumulates rapidly ($> 500\text{ m}$ in 60s).
- Mode B (Filtered IMU):** Incorporates running median and causal 20 Hz low-pass filtering. Suppresses engine acoustic noise spikes.

- **Mode C (AI Velocity Direct):** Directly integrates TCN-predicted 2D velocity vectors without Kalman filtering:

$$\mathbf{p}_k = \mathbf{p}_{k-1} + \mathbf{v}_{tcn} \Delta t$$
- **Mode D (AI + EKF):** Fuses TCN velocity as a pseudo-measurement into the 15-state EKF, actively estimating accelerometer and gyroscope biases.
- **Mode E (AI + EKF + NHC):** Activates Non-Holonomic Constraints ($v_y \approx 0, v_z \approx 0$), eliminating lateral vehicle slide and vertical altitude drift.
- **Mode F (AI + EKF + NHC + ZUPT):** Integrates zero-velocity updates during traffic stops, freezing position error accumulation during stationary intervals.
- **Mode G (Full System):** Snaps the filtered state trajectory to OpenStreetMap road centerlines, eliminating cumulative cross-track angular drift.

3. Evaluation Metrics & Mathematical Definitions

All positioning metrics are computed in horizontal meters relative to the local East-North-Up (ENU) tangent origin:

3.1 Absolute Trajectory Error (ATE RMSE)

Root Mean Squared Error between estimated coordinates $[p_E, p_N]$ and ground-truth reference coordinates $[p_E^{ref}, p_N^{ref}]$ over scorable timesteps N :

$$\text{ATE RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left((p_{E,k} - p_{E,k}^{ref})^2 + (p_{N,k} - p_{N,k}^{ref})^2 \right)}$$

3.2 Maximum & Final Position Drift

- **Maximum Drift:** $\max_k \|\mathbf{p}_k - \mathbf{p}_k^{ref}\|$ over the outage duration.
- **Final Position Drift:** Euclidean error at the concluding epoch of the outage window:

$$e_{final} = \|\mathbf{p}_{outage_end} - \mathbf{p}_{outage_end}^{ref}\|$$

3.3 Velocity Vector RMSE

$$\text{Velocity RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left((v_{E,k} - v_{E,k}^{ref})^2 + (v_{N,k} - v_{N,k}^{ref})^2 \right)}$$

3.4 Heading Angular Error (RMSE)

Wrapped angular error in degrees, computed strictly when vehicle ground speed exceeds 0.5 m/s to avoid stationary gimbal noise:

$$\Delta\psi_k = \text{wrap}_{180}(\psi_k - \psi_k^{ref})$$

$$\text{Heading RMSE} = \sqrt{\frac{1}{M} \sum_{j=1}^M (\Delta\psi_j)^2}$$

4. Execution Protocols

4.1 Running the Benchmark Suite

To execute the automated ablation suite against an input dataset:

```
# Replay full seven-stage ablation on a trip recording:
python replay.py --dataset data/raw_trips/held_out_trip.json --gnss-outage 60 --outage-start 10 --ablations

# Equivalent benchmark script:
python scripts/benchmark.py --dataset data/raw_trips/held_out_trip.json --gnss-outage 60
```

4.2 Output File Artifacts

The evaluation writes structured audit artifacts to *results/replay/*:

- *trip_results.json*: Full configuration parameters, component call counts, execution timings, and numerical metrics across whole-trip, outage, and recovery intervals.

- *trip_[A-G]_trajectory.csv*: Detailed timestamped state trajectories (\$t, p_E, p_N, v_E, v_N, \psi\$, mode, GNSS allowed).
- *trip_[A-G]_trajectory.json*: Interactive playback files renderable directly in the browser dashboard.
- *trip_ablation.csv*: Consolidated side-by-side comparison table of all modes.

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